



IoT End-to-End Big Data Streaming Pipeline

Kafka → Spark → PostgreSQL → Grafana

Naif Alqahtani 202172110

Seifeldin Gomaa 202045680

Mohammad Alhathal 202021080

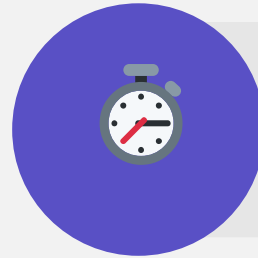




Project Objective



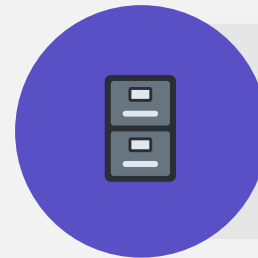
Simulate IoT sensor data (temperature & humidity)



Stream data in real time using Kafka



Process data using Spark Structured Streaming



Store processed data in PostgreSQL



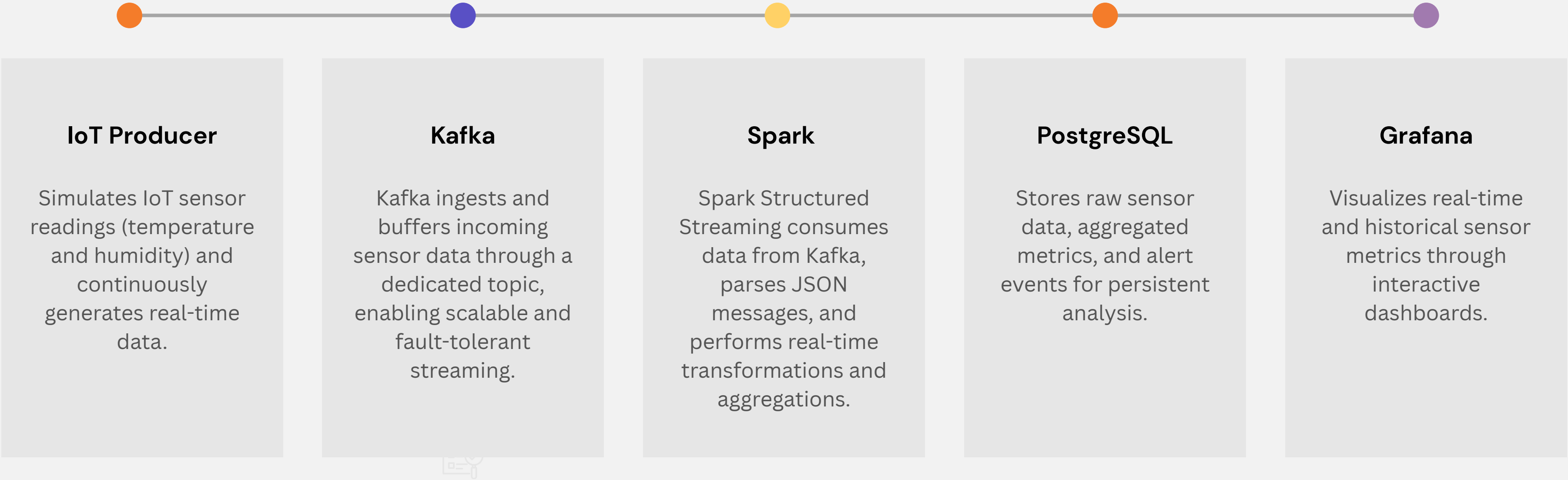
Visualize insights using Grafana





System Architecture

Data Flow

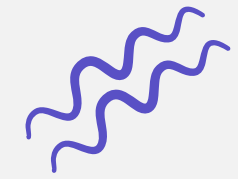


Technologies Used



- Apache Kafka – message streaming
- Apache Spark Structured Streaming – real-time processing
- PostgreSQL – persistent storage
- Grafana – dashboards & visualization
- Docker & Docker Compose – deployment

Data Ingestion & Kafka



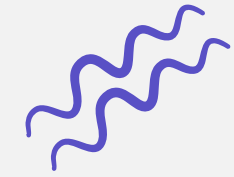
Why Kafka?

- High-throughput message ingestion
- Fault-tolerant and scalable
- Decouples data producers from consumers
- Supports real-time streaming use cases

Why not direct DB writes?

- Databases struggle with high write rates
- Kafka buffers spikes and prevents data loss

Data Ingestion & Kafka



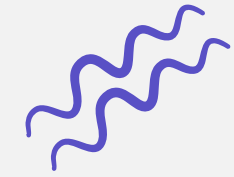
- Kafka & Zookeeper running :

```
PS D:\COLLEGE\last\ICS474\kafka-foundation> docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS        PORTS
af5a64de6d38   confluentinc/cp-kafka:7.5.3        "/etc/confluent/dock...  16 seconds ago Up 14 seconds 0.0.0.0:9092-
>9092/tcp, [::]:9092->9092/tcp   kafka
94e7d630f342   confluentinc/cp-zookeeper:7.5.3    "/etc/confluent/dock...  24 seconds ago Up 15 seconds 0.0.0.0:2181-
>2181/tcp, [::]:2181->2181/tcp   zookeeper
```

- Kafka topic `iot_sensors` created :

```
--replication-factor 1
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
Created topic iot_sensors.
PS D:\COLLEGE\last\ICS474\kafka-foundation> docker exec -it kafka kafka-topics --list --bootstrap-server localhost:9092
iot_sensors
PS D:\COLLEGE\last\ICS474\kafka-foundation> docker exec -it kafka kafka-topics --describe --topic iot_sensors --bootstrap-server localhost:9092
Topic: iot_sensors    TopicId: A7p99Nc7TnKFRwG-_DHYpg PartitionCount: 1      ReplicationFactor: 1    Configs:
    Topic: iot_sensors Partition: 0    Leader: 1      Replicas: 1      Isr: 1
```


Data Ingestion & Kafka



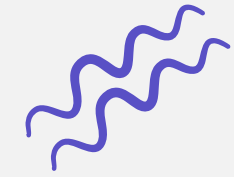
- Producer sending IoT data :

```
(.venv) PS D:\COLLEGE\last\ICS474\kafka-foundation> python .\producer\sensor_producer.py
Sending messages to topic 'iot_sensors' on localhost:9092 ... Press CTRL+C to stop.
D:\COLLEGE\last\ICS474\kafka-foundation\producer\sensor_producer.py:25: DeprecationWarning: datetime.datetime.utcnow()
al in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC)
  "timestamp": datetime.utcnow().isoformat(),
Sent: {'sensor_id': 'sensor-2', 'temperature': 31.51, 'humidity': 76.21, 'timestamp': '2025-12-17T19:02:53.017902'}
Sent: {'sensor_id': 'sensor-1', 'temperature': 32.22, 'humidity': 72.6, 'timestamp': '2025-12-17T19:02:54.164696'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 32.58, 'humidity': 72.03, 'timestamp': '2025-12-17T19:02:55.173864'}
Sent: {'sensor_id': 'sensor-1', 'temperature': 32.46, 'humidity': 66.42, 'timestamp': '2025-12-17T19:02:56.183249'}
Sent: {'sensor_id': 'sensor-2', 'temperature': 34.57, 'humidity': 61.95, 'timestamp': '2025-12-17T19:02:57.192097'}
Sent: {'sensor_id': 'sensor-1', 'temperature': 33.97, 'humidity': 67.88, 'timestamp': '2025-12-17T19:02:58.201127'}
Sent: {'sensor_id': 'sensor-2', 'temperature': 18.45, 'humidity': 35.55, 'timestamp': '2025-12-17T19:02:59.209781'}
Sent: {'sensor_id': 'sensor-2', 'temperature': 20.61, 'humidity': 67.97, 'timestamp': '2025-12-17T19:03:00.219543'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 28.22, 'humidity': 53.9, 'timestamp': '2025-12-17T19:03:01.229063'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 22.63, 'humidity': 66.59, 'timestamp': '2025-12-17T19:03:02.237471'}
Sent: {'sensor_id': 'sensor-1', 'temperature': 27.84, 'humidity': 55.52, 'timestamp': '2025-12-17T19:03:03.242428'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 29.21, 'humidity': 55.77, 'timestamp': '2025-12-17T19:03:04.251388'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 28.36, 'humidity': 47.45, 'timestamp': '2025-12-17T19:03:05.260639'}
Sent: {'sensor_id': 'sensor-3', 'temperature': 22.79, 'humidity': 47.24, 'timestamp': '2025-12-17T19:03:06.269044'}
Sent: {'sensor_id': 'sensor-1', 'temperature': 23.79, 'humidity': 73.83, 'timestamp': '2025-12-17T19:03:07.277626'}
```

- Consumer receiving streamed data :

```
PS D:\COLLEGE\last\ICS474\kafka-foundation> docker exec -it kafka kafka-console-consumer --bootstrap-server loc
{"sensor_id": "sensor-2", "temperature": 31.51, "humidity": 76.21, "timestamp": "2025-12-17T19:02:53.017902"}
{"sensor_id": "sensor-1", "temperature": 32.22, "humidity": 72.6, "timestamp": "2025-12-17T19:02:54.164696"}
{"sensor_id": "sensor-3", "temperature": 32.58, "humidity": 72.03, "timestamp": "2025-12-17T19:02:55.173864"}
{"sensor_id": "sensor-1", "temperature": 32.46, "humidity": 66.42, "timestamp": "2025-12-17T19:02:56.183249"}
{"sensor_id": "sensor-2", "temperature": 34.57, "humidity": 61.95, "timestamp": "2025-12-17T19:02:57.192097"}
{"sensor_id": "sensor-1", "temperature": 33.97, "humidity": 67.88, "timestamp": "2025-12-17T19:02:58.201127"}
{"sensor_id": "sensor-2", "temperature": 18.45, "humidity": 35.55, "timestamp": "2025-12-17T19:02:59.209781"}
{"sensor_id": "sensor-2", "temperature": 20.61, "humidity": 67.97, "timestamp": "2025-12-17T19:03:00.219543"}
{"sensor_id": "sensor-3", "temperature": 28.22, "humidity": 53.9, "timestamp": "2025-12-17T19:03:01.229063"}
{"sensor_id": "sensor-3", "temperature": 22.63, "humidity": 66.59, "timestamp": "2025-12-17T19:03:02.237471"}
{"sensor_id": "sensor-1", "temperature": 27.84, "humidity": 55.52, "timestamp": "2025-12-17T19:03:03.242428"}
{"sensor_id": "sensor-3", "temperature": 29.21, "humidity": 55.77, "timestamp": "2025-12-17T19:03:04.251388"}
{"sensor_id": "sensor-3", "temperature": 28.36, "humidity": 47.45, "timestamp": "2025-12-17T19:03:05.260639"}
{"sensor_id": "sensor-3", "temperature": 22.79, "humidity": 47.24, "timestamp": "2025-12-17T19:03:06.269044"}
{"sensor_id": "sensor-1", "temperature": 23.79, "humidity": 73.83, "timestamp": "2025-12-17T19:03:07.277626"}
{"sensor_id": "sensor-2", "temperature": 22.4, "humidity": 61.69, "timestamp": "2025-12-17T19:03:08.285999"}
{"sensor_id": "sensor-2", "temperature": 24.31, "humidity": 32.56, "timestamp": "2025-12-17T19:03:09.295226"}
```

Stream Processing with Spark



Why Spark for Streaming?

- Designed for distributed data processing
- Handles large-scale streams efficiently
- Fault-tolerant via checkpointing
- Easy Kafka integration

Stream Processing with Spark



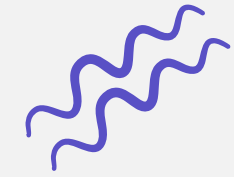
- View Spark Streaming Output :

```
spark
spark
spark Batch: 1066
spark
spark +-----+-----+-----+-----+
spark |sensor_id|temperature|humidity|event_time|
spark +-----+-----+-----+-----+
spark | sensor-3|      24.8|    66.16|2025-12-18 01:02:...|
spark +-----+-----+-----+-----+
spark
spark
spark Batch: 1067
spark
spark +-----+-----+-----+-----+
spark |sensor_id|temperature|humidity|event_time|
spark +-----+-----+-----+-----+
spark | sensor-2|      24.88|    75.57|2025-12-18 01:02:...|
spark +-----+-----+-----+-----+
```

- View Producer Logs :

```
PS C:\Users\Admin\Desktop\iot2\iot_pipeline_rebuild> docker-compose logs producer --tail 20
producer Sent: {'sensor_id': 'sensor-1', 'temperature': 23.4, 'humidity': 59.19, 'timestamp': '2025-12-18T01:02:39.216850'}
producer Sent: {'sensor_id': 'sensor-1', 'temperature': 18.47, 'humidity': 41.71, 'timestamp': '2025-12-18T01:02:40.222806'}
producer Sent: {'sensor_id': 'sensor-3', 'temperature': 31.26, 'humidity': 74.87, 'timestamp': '2025-12-18T01:02:41.226253'}
producer Sent: {'sensor_id': 'sensor-3', 'temperature': 25.62, 'humidity': 58.88, 'timestamp': '2025-12-18T01:02:42.228784'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 20.36, 'humidity': 52.8, 'timestamp': '2025-12-18T01:02:43.231749'}
producer Sent: {'sensor_id': 'sensor-3', 'temperature': 24.8, 'humidity': 66.16, 'timestamp': '2025-12-18T01:02:44.236057'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 24.88, 'humidity': 75.57, 'timestamp': '2025-12-18T01:02:45.238476'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 23.26, 'humidity': 51.16, 'timestamp': '2025-12-18T01:02:46.242153'}
producer Sent: {'sensor_id': 'sensor-1', 'temperature': 24.36, 'humidity': 40.06, 'timestamp': '2025-12-18T01:02:47.244406'}
producer Sent: {'sensor_id': 'sensor-1', 'temperature': 21.88, 'humidity': 73.1, 'timestamp': '2025-12-18T01:02:48.246730'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 20.01, 'humidity': 38.59, 'timestamp': '2025-12-18T01:02:49.249121'}
producer Sent: {'sensor_id': 'sensor-3', 'temperature': 33.74, 'humidity': 52.66, 'timestamp': '2025-12-18T01:02:50.255290'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 34.35, 'humidity': 59.65, 'timestamp': '2025-12-18T01:02:51.260555'}
producer Sent: {'sensor_id': 'sensor-3', 'temperature': 31.83, 'humidity': 38.82, 'timestamp': '2025-12-18T01:02:52.262832'}
producer Sent: {'sensor_id': 'sensor-1', 'temperature': 32.96, 'humidity': 40.89, 'timestamp': '2025-12-18T01:02:53.265198'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 34.52, 'humidity': 36.14, 'timestamp': '2025-12-18T01:02:54.270472'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 22.03, 'humidity': 31.29, 'timestamp': '2025-12-18T01:02:55.273633'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 33.8, 'humidity': 58.99, 'timestamp': '2025-12-18T01:02:56.277182'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 30.08, 'humidity': 51.33, 'timestamp': '2025-12-18T01:02:57.282837'}
producer Sent: {'sensor_id': 'sensor-2', 'temperature': 18.3, 'humidity': 34.37, 'timestamp': '2025-12-18T01:02:58.288109'}
PS C:\Users\Admin\Desktop\iot2\iot_pipeline_rebuild>
```

Stream Processing with Spark



- Verify Kafka Topic Messages :

```
PS C:\Users\Admin\Desktop\iot2\iot_pipeline_rebuild> docker exec kafka /opt/kafka/bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic iot_sensors --from-beginning --max-messages 5
{"sensor_id": "sensor-2", "temperature": 24.48, "humidity": 70.82, "timestamp": "2025-12-18T00:37:13.610233"}
{"sensor_id": "sensor-2", "temperature": 31.49, "humidity": 59.06, "timestamp": "2025-12-18T00:37:14.834106"}
{"sensor_id": "sensor-3", "temperature": 25.17, "humidity": 72.09, "timestamp": "2025-12-18T00:37:15.840236"}
{"sensor_id": "sensor-1", "temperature": 23.52, "humidity": 59.39, "timestamp": "2025-12-18T00:37:16.844959"}
{"sensor_id": "sensor-3", "temperature": 29.57, "humidity": 72.57, "timestamp": "2025-12-18T00:37:17.851596"}
Processed a total of 5 messages
PS C:\Users\Admin\Desktop\iot2\iot_pipeline_rebuild>
```

- Monitor Real-time Streaming :

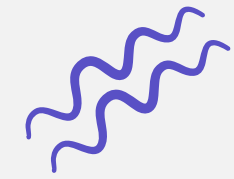
```
spark
> spark Batch: 1172
spark
spark
+-----+-----+-----+-----+
|sensor_id|temperature|humidity|event_time|
+-----+-----+-----+-----+
| sensor-3|      22.33|    65.62|2025-12-18 01:04:...|
+-----+-----+-----+-----+

spark
Batch: 1173
spark
spark
+-----+-----+-----+-----+
|sensor_id|temperature|humidity|event_time|
+-----+-----+-----+-----+
| sensor-2|      25.01|    62.12|2025-12-18 01:04:...|
+-----+-----+-----+-----+

spark
Batch: 1174
spark
spark
+-----+-----+-----+-----+
|sensor_id|temperature|humidity|event_time|
+-----+-----+-----+-----+
| sensor-2|      32.1|    32.75|2025-12-18 01:04:...|
+-----+-----+-----+-----+

spark
spark
```

Storage & Visualization



PostgreSQL

- Reliable structured storage
- Supports time-based queries and aggregations
- Integrates well with Spark

Grafana

- Real-time dashboards
- Time-series visualization
- Alert monitoring

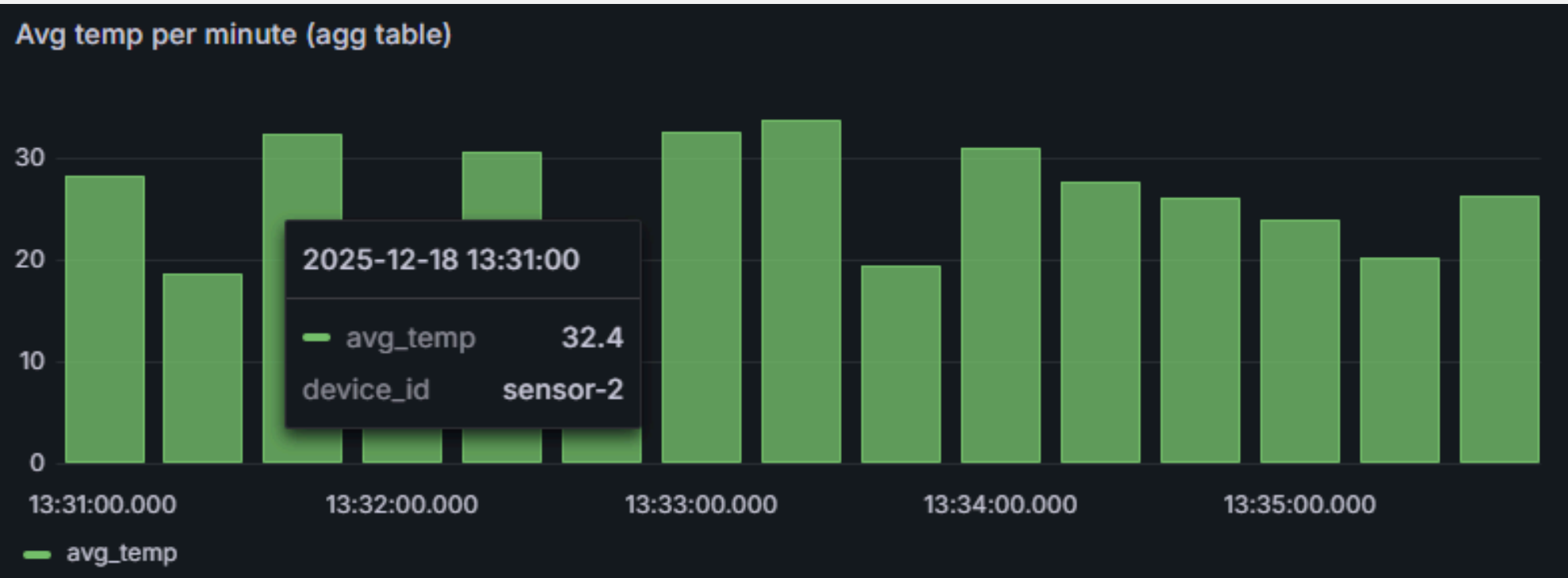
Storage & Visualization



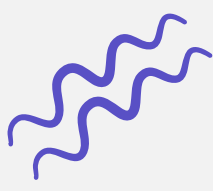
- Alerts list :

Alerts list				
time	device_id	rule	value	threshold
2025-12-18 13:35:10	sensor-2	temp_gt_threshold	35.0	32
2025-12-18 13:35:06	sensor-3	temp_gt_threshold	34.2	32
2025-12-18 13:35:05	sensor-1	temp_gt_threshold	33.1	32
2025-12-18 13:35:01	sensor-2	temp_gt_threshold	32.3	32
2025-12-18 13:34:58	sensor-1	temp_gt_threshold	33.8	32
2025-12-18 13:34:42	sensor-2	temp_gt_threshold	34.6	32

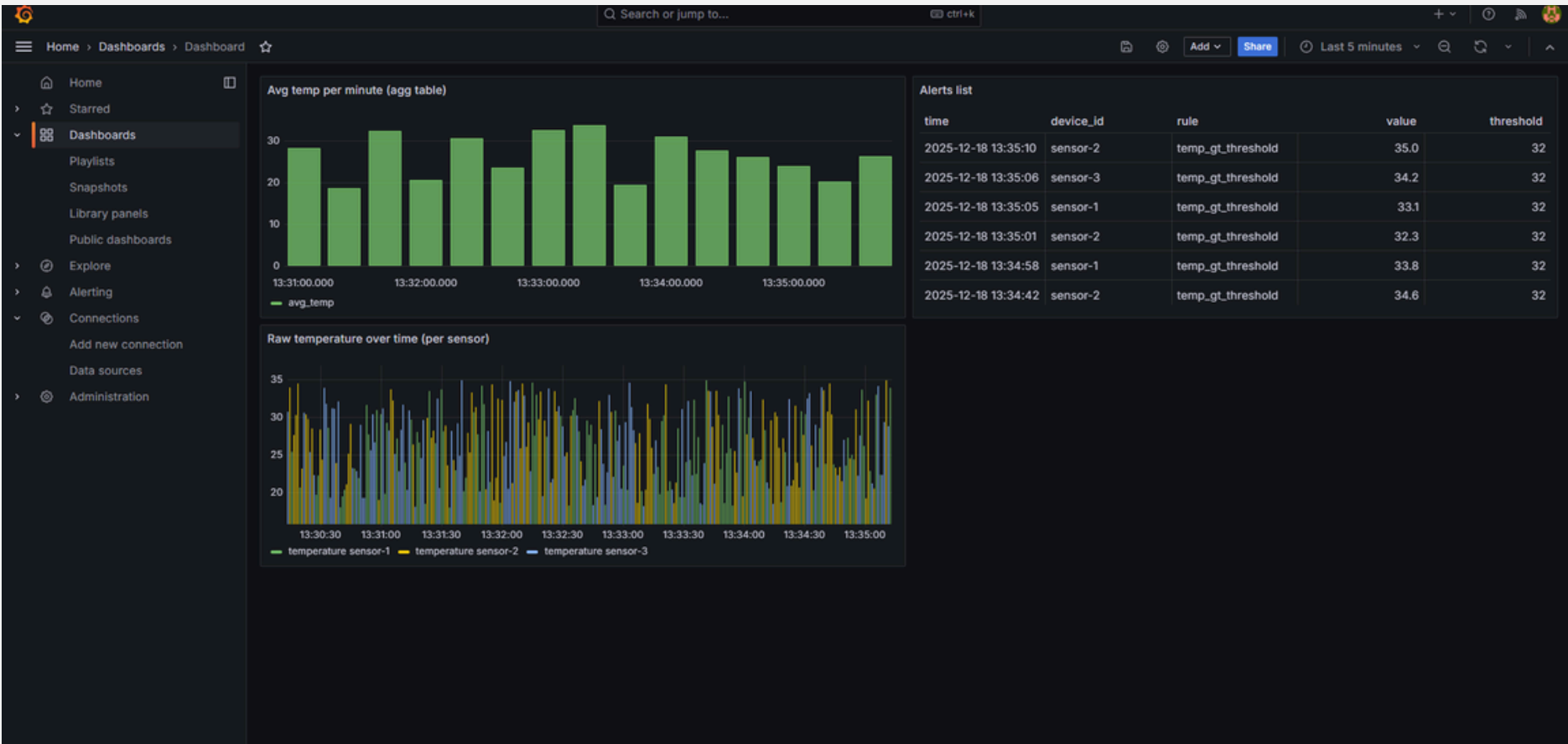
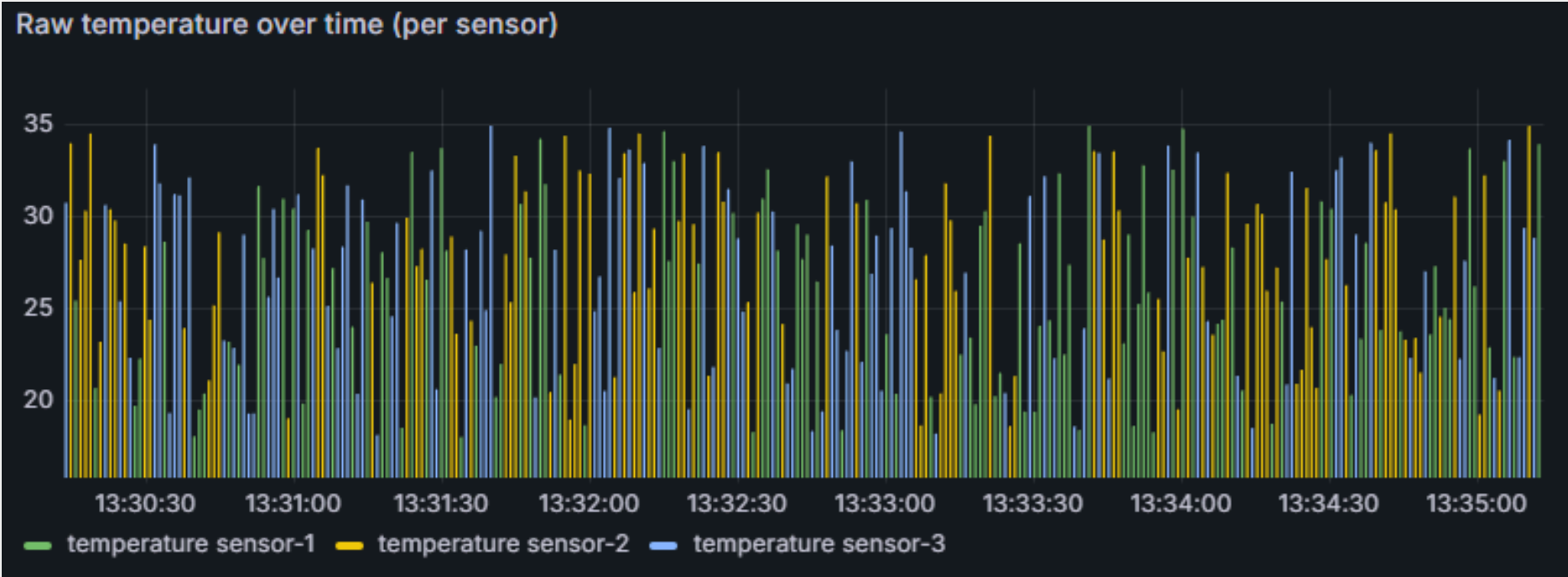
- Aggregate table:



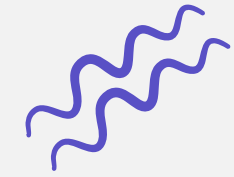
Storage & Visualization



- Raw temperature over time :
- Dashboard :



Key Learning Outcomes



**Real-time data
ingestion with Kafka**

**Stream processing
using Spark**

**Decoupling ingestion,
processing, and
storage**

**Handling high-
throughput IoT data**

**Building reproducible
pipelines with Docker**

**Real-time
visualization**

Challenges & Solutions



Challenges :

Kafka configuration

**Streaming data
reliability**

**Spark-Kafka
integration**

Solutions :

Dockerized services

**Topic-based
decoupling**

Schema enforcement

Conclusion



This project successfully demonstrates an end-to-end IoT streaming pipeline using Apache Kafka, Spark Structured Streaming, PostgreSQL, and Grafana.

Real-time sensor data is ingested, processed, stored, and visualized in a scalable and decoupled architecture.

The pipeline supports raw data storage, time-based aggregations, alert generation, and real-time dashboards, fulfilling all project requirements.

Overall, the system highlights the effectiveness of streaming architectures for handling continuous IoT data.



Q&A

